

A Comprehensive Multimodal Approach for Knee Osteoarthritis Diagnosis Integrating Radiographic and Clinical Data

Abstract—Knee osteoarthritis (OA) is a prevalent condition that demands accurate assessment for early diagnosis and effective management. This study presents a multimodal approach combining X-ray imaging and clinical data from the Osteoarthritis Initiative (OAI) to predict OA severity according to the Kellgren-Lawrence (KL) grading system. EfficientNetV2, selected for its superior performance and efficiency over models like ResNet and DenseNet, is utilized for feature extraction from X-ray images, while clinical features are processed through traditional machine learning techniques. The combined model achieves a prediction accuracy of 96% in soft tissue assessment areas. Data preprocessing steps, including resizing, normalization, augmentation, and encoding, ensure consistency across modalities. Results demonstrate that integrating imaging and clinical data significantly outperforms single-modality approaches. However, the model faces challenges in detecting subtle structural changes, such as early Joint Space Narrowing (JSN), which are critical for precise grading. This work highlights the potential of multimodal methods to enhance diagnostic accuracy and support personalized treatment planning for OA patients.

Keywords—Multimodal approach, Knee osteoarthritis, EfficientNetV2, Clinical data integration, X-ray imaging, Kellgren-Lawrence grade, Feature fusion, Medical diagnosis accuracy.

I. INTRODUCTION

Knee osteoarthritis (OA) is one of the leading causes of disability worldwide, significantly impairing joint function and quality of life. The disease is characterized by progressive degeneration of the cartilage, resulting in joint pain, stiffness, and reduced mobility [1] [2]. As the most common type of arthritis, knee OA affects millions of individuals globally, particularly the elderly population [3]. Its clinical manifestations include joint space narrowing (JSN), osteophyte formation, and subchondral bone changes, all of which are visible through radiographic imaging, particularly X-rays [4]. Among the various radiographic grading systems, the Kellgren-Lawrence (KL) grading scale is the most widely used, assessing OA severity based on radiographic evidence, ranging from 0 (no disease) to 4 (severe disease) [5].

The Kellgren-Lawrence grade, while widely used, is not without limitations. It primarily focuses on a few imaging features, namely osteophytes and joint space narrowing, and does not fully account for other critical OA features such as pain severity, subchondral sclerosis, and functional limitations, which are important for a more complete understanding of the disease [6]. Moreover, the KL grading system fails to represent the heterogeneity of OA, where patients may exhibit different patterns of joint damage that are not always captured by the

standard radiographic markers [7]. This gap in the diagnostic approach highlights the need for more advanced models that can assess multiple OA features simultaneously and provide a more comprehensive evaluation of disease severity.

In recent years, deep learning models, particularly convolutional neural networks (CNNs), have shown great promise in automating the diagnosis of knee OA from X-ray images. These models leverage CNNs to extract complex patterns from the images, enabling more accurate assessments of OA severity than traditional manual methods [4], [8], [9]. However, most of the existing models focus on single diagnostic tasks, such as predicting the KL grade or detecting osteophytes. Few models have integrated multiple OA features into a single framework [10], and even fewer combine both imaging data and clinical information, such as pain levels, age, and side of the knee, for a more holistic disease assessment [11].

This study aims to address these limitations by developing a multitask deep learning model that simultaneously predicts multiple knee OA features using both X-ray images and clinical data. The primary focus is on predicting the KL grade, osteophyte presence and severity, joint space narrowing (JSN), and pain intensity. The multimodal nature of this model allows it to combine radiographic evidence with clinical information to provide a more accurate and comprehensive prediction of OA severity, which can facilitate timely intervention and personalized treatment [12], [13].

The main contributions of this study are as follows:

- 1) Proposes a multitask deep learning model to predict key knee OA features, such as KL grade, osteophytes, JSN, and pain intensity, using both X-ray images and clinical data.
- 2) Combines X-ray images with clinical data like pain severity and demographics to improve OA diagnosis accuracy.
- 3) Provides a comprehensive understanding of knee OA by considering both visual features and clinical context, leading to better severity grading and decision-making.

This approach not only improves the prediction accuracy for knee OA but also offers a pathway for integrating multimodal data in clinical applications, ultimately leading to better patient outcomes and more personalized care strategies [7], [14]–[16].

The remainder of the paper is structured as follows: Section II reviews related work in this field, Section III outlines the methodology used, Section IV presents the results of our suggested approach, and Section V concludes the study.

II. LITERATURE REVIEW

Knee osteoarthritis (OA) is a leading cause of chronic disability, primarily affecting the aging population. Characterized by the degradation of cartilage and joint structures, OA significantly impairs mobility and quality of life. Recent research recognizes OA as a multifactorial condition, involving not only cartilage but also bone, synovium, and muscles, complicating both diagnosis and management. Advanced imaging techniques, particularly magnetic resonance imaging (MRI), have significantly improved our ability to assess joint abnormalities. MRI can capture cartilage loss, bone marrow lesions (BMLs), joint effusion, and synovitis, which correlate with disease progression and symptom severity [1], [3]. However, the relationship between structural changes and pain remains inconsistent, with some patients experiencing significant structural damage without corresponding pain, while others report severe pain despite limited damage [5], [17].

Knee OA is increasingly viewed as a heterogeneous disease influenced by genetic, mechanical, and environmental factors. This variability complicates the identification of reliable biomarkers for diagnosis and treatment. Recent studies have expanded the focus beyond cartilage loss to include the role of subchondral bone and synovium in disease progression and pain [6], [7]. To better understand the disease, comprehensive diagnostic models integrating structural, biochemical, and clinical data are needed.

Machine learning (ML) models, especially convolutional neural networks (CNNs), have demonstrated promising results in classifying knee OA severity based on imaging data. These models can detect early-stage degeneration, such as cartilage loss and BMLs, which are often missed by traditional methods. ML techniques have also been shown to predict pain severity and functional impairment, providing new insights into the link between structural changes and clinical outcomes [4], [10]. Despite their potential, the application of ML in OA research faces challenges related to the availability of high-quality, diverse datasets and the interpretability of complex models. Many existing studies rely on small or homogeneous datasets, limiting model generalizability and real-world applicability [11], [12]. Additionally, the "black-box" nature of deep learning models restricts their clinical adoption, as healthcare providers need to understand how models arrive at predictions to trust their decisions [13], [14].

Integrating non-imaging data, such as clinical assessments, genetic markers, and inflammatory biomarkers, alongside imaging data, can enhance ML models for knee OA. Combining these data types has shown to improve prediction accuracy, especially for identifying patients at risk of rapid disease progression [15], [16]. Genetic research has identified key genes involved in cartilage metabolism, and incorporating this genomic data into ML models could help predict individuals predisposed to severe OA, even before clinical symptoms appear [18]. Recent advancements in **multitask deep learning models** further highlight the integration of multimodal imaging data, which can predict OA features such as joint space

narrowing and osteophytes, paving the way for more effective early diagnoses and personalized treatment plans [9].

Table I summarizes recent research related to knee osteoarthritis (OA) diagnosis and severity prediction.

TABLE I: Summary of Related Work

Paper	Author's	Dataset	Methodology	Research Gap
[1]	Zhengkuan Zhao, Mingkuan Zhao, Tao Yang, Jie Li, Chao Qin, Ben Wang, Li Wang, Bing Li, Jun Liu	Osteoarthritis Initiative (OAI) dataset	ML models (Random Forests, SVM, CNN, CAM)	Limited focus on imaging data, needs clinical data integration for better prediction of pain severity
[3]	Jiaping Hu, Chuanyang Zheng, Qingling Yu, et al.	OAI dataset, MRI data	3D DenseNet169 for OA progression prediction	Dataset size limits generalizability; no clinical data integration
[17]	Liangliang Liu, Jing Chang, Pei Zhang, et al.	OAI dataset, Clinical + Imaging data	Multi-Modal Learning Method (MMLM), XGBoost, ResNet50	Lack of a robust fusion method for multimodal data, could improve accuracy
[4]	Aleksei Tiulpin, et al.	Radiographic data, clinical data	CNN + Gradient Boosting Machine (GBM), multimodal learning	Could improve KL-grade prediction and total knee replacement forecasting
[9]	Yun Xin Teoh, Alice Othmani, Khin Wee Lai, Siew Li Goh, Juliana Usman	OAI dataset, Multimodal imaging data	CNN + KNN classifier, multitask deep learning	Needs further testing on diverse datasets to validate generalizability

III. MATERIALS AND METHODS

The methodology outlines a multimodal approach for predicting knee osteoarthritis (OA) severity. In the Data Collection subsection, X-ray images and clinical tabular data are gathered. During Preprocessing, images are resized, normalized, and augmented, while clinical data is encoded, imputed, and feature-selected. The Data Integration and Splitting subsection combines the datasets and splits them into training and testing sets. The Algorithm subsection describes using EfficientNetV2 for X-ray feature extraction for clinical data analysis to predict the Kellgren-Lawrence (KL) grade. Model evaluation is conducted using metrics like accuracy, precision, recall, and F1-score. Figure 1 illustrates the workflow.

A. Data collection

For this study, two datasets were utilized: one consisting of **X-ray images** and the other containing **clinical tabular data**. Both datasets were collected from Kaggle and used together to predict the severity of knee osteoarthritis (OA) based on the Kellgren-Lawrence (KL) grading system.

The **X-ray dataset** includes knee joint images used for the detection of osteoarthritis and its grading using the KL scale. The **CSV dataset** provides clinical and demographic data about the patients, including osteophytes, joint space narrowing (JSN), and functional limitations. A comprehensive summary of the dataset and descriptions of each column are given in Table I and Table II.

TABLE II: X-ray Dataset Description

Grade	Description
0	Healthy knee
1	Doubtful joint narrowing with possible osteophytic lipping
2	Definite osteophytes and possible joint space narrowing
3	Multiple osteophytes, definite joint space narrowing, mild sclerosis
4	Large osteophytes, significant joint narrowing, severe sclerosis

TABLE III: CSV Dataset Description

Attribute	Description
id	Unique identifier for each patient
side	Specifies whether the knee is left (L) or right (R)
subset	Indicates whether the sample is in the train, validation, or test set
filename	Name of the associated X-ray image file
kl_grade	Target variable representing the KL grade
osteophytes	Presence of osteophytes
jsn	Joint space narrowing (JSN)
osfl	Functional limitations observed at various time-points

B. Data Preprocessing

In this study data preprocessing was conducted in two stages: one for the **X-ray images** and the other for the **clinical CSV data**.

1. X-ray Image Preprocessing:

- **Image Resizing:** X-ray images were resized to 224x224 pixels to ensure a uniform input size.
- **Color Adjustment:** Images with inverted color schemes were corrected using a custom function that checks the average pixel values at different image regions.
- **Normalization:** Pixel values were normalized to a range of [0, 1] for standardization.
- **Augmentation:** Data augmentation techniques such as rotation, flipping, and cropping were applied to enhance the dataset and reduce overfitting.

2. CSV Data Preprocessing:

- **Label Encoding:** The `kl_grade` and other clinical labels were encoded using `OneHotEncoding`. Missing values were handled by filling them with default values.
- **Feature Selection:** Relevant clinical features like osteophytes, joint space narrowing, and others were selected, while irrelevant columns were removed.
- **Handling Missing Data:** Missing clinical data were handled by imputation to ensure no NaN values remained.

C. Data Integration and Splitting

The datasets consisting of X-ray images and clinical data were first combined to create a unified multimodal dataset. The X-ray images provided visual evidence of knee osteoarthritis,

while the clinical data offered additional demographic and medical information. These datasets were merged based on a common identifier (patient ID or filename), ensuring that each sample included both imaging and clinical features.

Following data integration, the combined dataset was split into training and testing sets with an 80-20 ratio. Specifically, 80% of the data was allocated to the training set, while the remaining 20% was set aside for testing and model evaluation. This approach ensured sufficient training data for the model while maintaining a separate test set to assess its performance.

D. Multimodal Knee OA Prediction Algorithm

The multimodal knee osteoarthritis severity prediction provided in Algorithm 1 - integrates X-ray images and clinical data to predict Kellgren-Lawrence (KL) grades. It preprocesses image and tabular data, extracts features using EfficientNetV2, and combines them for classification. The algorithm is evaluated using accuracy, precision, recall, and F1-score.

Algorithm 1 Multimodal Knee Osteoarthritis Severity Prediction Algorithm

X-ray image data X , Clinical data C Predicted KL Grade y_i

Step 1: Data Collection

Collect X-ray images X and clinical data C .

Step 2: Data Preprocessing

each image $x_i \in X$ Process image using resizing, normalization, and augmentation:

$x_i \rightarrow \frac{x_i - \mu_x}{\sigma_x}$ each clinical data entry $c_i \in C$ Encode categorical features, impute missing values:

$c_i \rightarrow \hat{c}_i$

Step 3: Data Integration

Merge X and C based on common identifier ID :

$D = \{(x_i, c_i)\}$

Step 4: Data Splitting

Split dataset into training and testing sets with 80%-20% ratio:

$D_{\text{train}} = 0.8 \times D, \quad D_{\text{test}} = 0.2 \times D$

Step 5: Model Training

Train EfficientNetV2 model on X-ray data to extract image features:

$F_{\text{image}} = f_{\text{EffNetV2}}(X)$

Train machine learning model on clinical data:

$F_{\text{clinical}} = f_{\text{ML}}(C)$

Step 6: Feature Fusion

Concatenate features from both modalities:

$F_{\text{combined}} = [F_{\text{image}}, F_{\text{clinical}}]$

Step 7: Model Evaluation

Evaluate the model on the test set using metrics (accuracy, precision, recall):

$y_i = \text{Classifier}(F_{\text{combined}})$

Step 8: Prediction

Predict KL grade for unseen data $(x_{\text{new}}, c_{\text{new}})$:

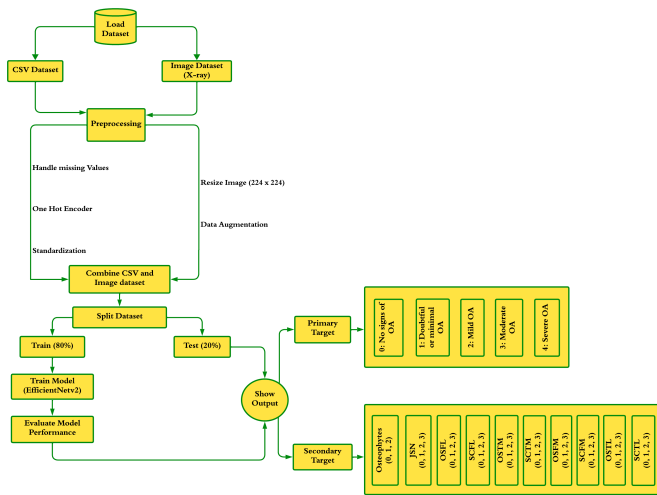
$F_{\text{new}} = [f_{\text{EffNetV2}}(x_{\text{new}}), f_{\text{ML}}(c_{\text{new}})]$

$y_{\text{new}} = \text{Classifier}(F_{\text{new}})$

Here, X represents the set of X-ray images, and C represents the corresponding clinical data entries. Each $x_i \in X$ is

an individual X-ray image, and $c_i \in C$ is its paired clinical data. F_{image} and F_{clinical} denote the features extracted from the images and clinical data, respectively. These features are combined into F_{combined} , which serves as the input for the classifier to predict the KL grade y_i , where $y_i \in \{0, 1, 2, 3, 4\}$.

E. Multimodal Knee OA Severity Prediction Workflow



After preprocessing, the X-ray and clinical datasets are integrated into a unified multimodal dataset, aligning the data based on patient identifiers. This integrated dataset is then split into training and testing sets using an 80-20 ratio, with the training set used to train the model and the testing set used for performance evaluation.

The fused features are subsequently input into a classifier, which predicts the Kellgren-Lawrence (KL) grade, a standardized measure of OA severity. The classifier is trained using the training set and evaluated on the testing set using various

Finally, to predict the KL grade for new, unseen data, the trained model extracts features from the input X-ray images and clinical data, fuses them, and applies the classifier to generate the predicted KL grade. By combining both imaging and clinical data, this workflow aims to enhance the accuracy, robustness, and clinical applicability of OA severity prediction, contributing to improved early diagnosis and personalized treatment planning.

Figure 2 illustrates the architecture of the neural network utilized in this study, designed for feature extraction and classification. The structure integrates multiple computational layers and modules to process input data effectively.

Fig. 2: Neural Network Architecture

(1536), enhance feature extraction at different levels of abstraction.

- **Batch Normalization and Activation:** Layers like BatchNorm2D and Conv2DNormActivation (1280) stabilize the training process and prevent overfitting by normalizing activations.
- **Squeeze and Excitation (SE) Module:** The Squeeze-and-Excitation block recalibrates channel-wise feature responses to enhance the network’s representational capacity.
- **Pooling Layers:** Adaptive average pooling layers aggregate features across spatial dimensions, ensuring consistent output sizes irrespective of input dimensions.
- **Classifier:** A fully connected linear layer (1280 → 44) maps the extracted features to the output space.
- **Output Layer:** The final layer applies a sigmoid activation function for classification, outputting probabilities for the target classes.

The network structure emphasizes modularity and efficiency by integrating advanced layers such as MBConv and SE blocks while maintaining a clear flow from input to output. The use of dotted lines signifies interactions and dependencies between the layers, illustrating the architectural complexity and multi-path feature flow. This framework supports robust feature extraction and classification, making it highly suited for tasks requiring multimodal data processing. ““

IV. RESULTS AND DISCUSSIONS

This section provides an overview of the experimental results and discusses the observations with a comparative analysis between SL and SSL techniques for PCOS detection.

A. Performance Measures

The performance of the multimodal knee osteoarthritis (OA) severity prediction model was evaluated using several classification metrics: **accuracy**, **precision**, **recall** and **F1-score**. Accuracy measures the proportion of correct predictions among all predictions. Precision indicates the proportion of true positive predictions out of all positive predictions, while recall focuses on the proportion of true positives among all actual positive cases. The F1-score is the harmonic mean of precision and recall, balancing the two metrics. These metrics collectively provide a comprehensive evaluation of the model’s effectiveness in predicting OA severity.

B. Training Performance Analysis

Figure 3 illustrates the accuracy and loss per epoch for both the training and validation datasets for over 30 epochs. It shows a steady increase in validation accuracy, indicating its improved ability to classify knee osteoarthritis severity over time. Additionally, the decreasing validation loss further confirms that the model is learning the underlying patterns in the data without overfitting.

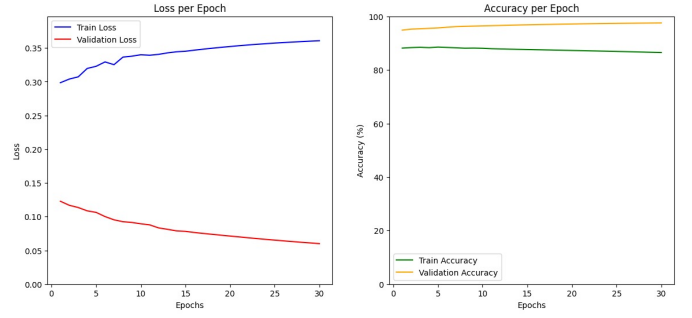


Fig. 3: Accuracy and Loss curve

C. Comparative Performance Evaluation

The evaluation of the performance across the various sectors of knee osteoarthritis (OA) is summarized in Table IV and visualized in Figure 4 and 5. Table IV compares performance across 11 OA sectors, with SCFL/SCTL achieving 96% accuracy and JSN showing the lowest recall at 22%.

TABLE IV: Comparative Performance Metrics

Sector	Accuracy	Precision	Recall	F1-Score
KL_GRADE	0.6848	0.75	0.82	0.78
OSTEOPHYTES	0.6836	0.77	0.85	0.81
JSN	0.7609	0.49	0.22	0.30
OSFL	0.6815	0.76	0.90	0.83
SCFL	0.9596	0.98	0.99	0.99
OSTM	0.6893	0.78	0.67	0.72
SCTM	0.8166	0.89	0.93	0.91
OSFM	0.6982	0.79	0.91	0.85
SCFM	0.8195	0.90	0.94	0.92
OSTL	0.7584	0.84	0.92	0.88
SCTL	0.9586	0.99	0.99	0.99

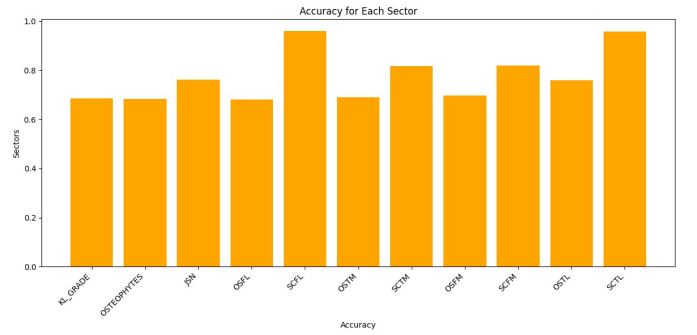


Fig. 4: Accuracy Comparison

Figure 4 and 5 complement the table by visually representing the model’s performance across the sectors. **Figure 4** (Accuracy Comparison) highlights the significant differences in accuracy, with soft tissue sectors like SCFL and SCTL achieving much higher accuracy compared to structural sectors like JSN, likely due to better-defined features.

Figure 5 (Precision, Recall, and F1-Score Comparison) illustrates the balance between precision, recall, and F1-score, showcasing the strong performance of SCFL and SCTL in minimizing false positives, while the JSN sector struggles with

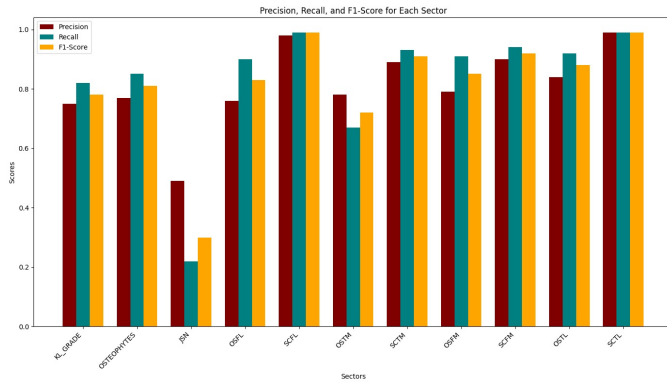


Fig. 5: Precision, Recall, and F1-Score Comparison

a notable discrepancy between precision and recall, indicating challenges in detecting joint space narrowing.

The performance of the multimodal knee osteoarthritis (OA) severity prediction model demonstrates notable variations across different sectors. Soft tissue sectors, such as SCFL and SCTL, achieve the highest accuracy at approximately 96%, reflecting the model's strong ability to classify well-defined features. On the other hand, structural sectors like JSN show lower accuracy (around 76%) and a particularly low recall of 22%, indicating difficulty in detecting subtle changes in joint space. The KL-Grade and OSTEOPHYTES sectors also exhibit moderate accuracy (68%), suggesting challenges in distinguishing these features. Overall, the model performs well in certain sectors, but further refinement is needed for accurate detection in more complex areas.

V. CONCLUSION

The multimodal knee osteoarthritis (OA) severity prediction model successfully integrates X-ray imaging and OAI clinical data to predict the Kellgren-Lawrence (KL) grade, a key measure of OA severity. The methodology employs EfficientNetV2 for feature extraction from X-ray images, while machine learning models process clinical data. After data preprocessing, integration, and splitting into training and testing sets, the model is trained and evaluated on several metrics such as accuracy, precision, recall, F1-score, and confusion matrix. The results show promising performance, with soft tissue sectors like SCFL and SCTL achieving over 95% accuracy, while structural sectors such as joint space narrowing (JSN) presented challenges, including lower recall. Despite the strong performance, particularly in soft tissue areas, the model's ability to detect subtle structural changes in OA requires improvement. Future work should focus on refining the model for more accurate detection in complex sectors like JSN and KL grade, enhancing its robustness and clinical applicability for OA diagnosis and treatment planning.

REFERENCES

[1] Z. Zhao, M. Zhao, T. Yang, J. Li, C. Qin, B. Wang, L. Wang, B. Li, and J. Liu, "Identifying significant structural factors associated with knee pain severity in patients with osteoarthritis using machine learning," *Scientific Reports*, vol. 14, no. 1, p. 14705, 2024.

[2] M. Mahmood, T. Alam, and Z. Sultana, "Deepad-insight: A benchmark evaluation of deep learning architectures for multiclass alzheimer's disease classification and staging from brain mri data," in *2024 International Conference on Innovations in Science, Engineering and Technology (ICISSET)*. IEEE, 2024, pp. 1–6.

[3] J. Hu, C. Zheng, Q. Yu, L. Zhong, K. Yu, Y. Chen, Z. Wang, B. Zhang, Q. Dou, and X. Zhang, "Deepkoa: a deep-learning model for predicting progression in knee osteoarthritis using multimodal magnetic resonance images from the osteoarthritis initiative," *Quantitative Imaging in Medicine and Surgery*, vol. 13, no. 8, p. 4852, 2023.

[4] A. Tiulpin, S. Klein, S. M. Bierma-Zeinstra, J. Thevenot, E. Rahtu, J. v. Meurs, E. H. Oei, and S. Saarakkala, "Multimodal machine learning-based knee osteoarthritis progression prediction from plain radiographs and clinical data," *Scientific reports*, vol. 9, no. 1, p. 20038, 2019.

[5] X. Y. Teh, P. S. Q. Yeoh, T. Wang, X. Wu, K. Hasikin, and K. W. Lai, "Knee osteoarthritis diagnosis with unimodal and multi-modal neural networks: Data from the osteoarthritis initiative," *IEEE Access*, 2024.

[6] C. L. Piccolo, C. A. Mallio, F. Vaccarino, R. F. Grasso, and B. B. Zobel, "Imaging of knee osteoarthritis: a review of multimodal diagnostic approach," *Quantitative Imaging in Medicine and Surgery*, vol. 13, no. 11, p. 7582, 2023.

[7] C. Kokkoti, S. Moustakidis, E. Papageorgiou, G. Giakas, and D. Tsaopoulos, "Machine learning in knee osteoarthritis: A review," *Osteoarthritis and Cartilage Open*, vol. 2, no. 3, p. 100069, 2020.

[8] W. Cao, Y. Wu, C. Huang, M. J. Patwary, and X. Wang, "Mff: Multi-modal feature fusion for zero-shot learning," *Neurocomputing*, vol. 510, pp. 172–180, 2022.

[9] Y. X. Teoh, A. Othmani, K. W. Lai, S. L. Goh, and J. Usman, "Stratifying knee osteoarthritis features through multitask deep hybrid learning: data from the osteoarthritis initiative," *Computer Methods and Programs in Biomedicine*, vol. 242, p. 107807, 2023.

[10] M. A. Khan, S. Kadry, P. Parwekar, R. Damaševičius, A. Mehmood, J. A. Khan, and S. R. Naqvi, "Human gait analysis for osteoarthritis prediction: A framework of deep learning and kernel extreme learning machine," *Complex & Intelligent Systems*, pp. 1–19, 2021.

[11] J.-B. Schiratti, R. Dubois, P. Herent, D. Cahané, J. Dachary, T. Clozel, G. Wainrib, F. Keime-Guibert, A. Lalande, M. Pueyo *et al.*, "A deep learning method for predicting knee osteoarthritis radiographic progression from mri," *Arthritis Research & Therapy*, vol. 23, pp. 1–10, 2021.

[12] S. Rani, M. Memoria, A. Almogren, S. Bharany, K. Joshi, A. Altameem, A. U. Rehman, and H. Hamam, "Deep learning to combat knee osteoarthritis and severity assessment by using cnn-based classification," *BMC Musculoskeletal Disorders*, vol. 25, no. 1, p. 817, 2024.

[13] J. Martel-Pelletier, P. Palement, and J.-P. Pelletier, "Magnetic resonance imaging assessments for knee segmentation and their use in combination with machine/deep learning as predictors of early osteoarthritis diagnosis and prognosis," *Therapeutic Advances in Musculoskeletal Disease*, vol. 15, p. 1759720X231165560, 2023.

[14] E. Panfilov, S. Saarakkala, M. T. Nieminen, and A. Tiulpin, "End-to-end prediction of knee osteoarthritis progression with multi-modal transformers," *arXiv preprint arXiv:2307.00873*, 2023.

[15] K. Leung, B. Zhang, J. Tan, Y. Shen, K. J. Geras, J. S. Babb, K. Cho, G. Chang, and C. M. Deniz, "Prediction of total knee replacement and diagnosis of osteoarthritis by using deep learning on knee radiographs: data from the osteoarthritis initiative," *Radiology*, vol. 296, no. 3, pp. 584–593, 2020.

[16] Y. X. Teoh, K. W. Lai, J. Usman, S. L. Goh, H. Mohafez, K. Hasikin, P. Qian, Y. Jiang, Y. Zhang, and S. Dhanalakshmi, "[retracted] discovering knee osteoarthritis imaging features for diagnosis and prognosis: Review of manual imaging grading and machine learning approaches," *Journal of healthcare engineering*, vol. 2022, no. 1, p. 4138666, 2022.

[17] L. Liu, J. Chang, P. Zhang, Q. Ma, H. Zhang, T. Sun, and H. Qiao, "A joint multi-modal learning method for early-stage knee osteoarthritis disease classification," *Heliyon*, vol. 9, no. 4, 2023.

[18] C. Kokkoti, S. Moustakidis, G. Giakas, and D. Tsaopoulos, "Identification of risk factors and machine learning-based prediction models for knee osteoarthritis patients," *Applied Sciences*, vol. 10, no. 19, p. 6797, 2020.